PitchProfile — Model Ownership & Mastery Guide

Oct 22, 2026 · @hadi

Five model-ownership groups, built so every member can present one and field questions on it. Team self-selects who takes which group.

How this guide works

Five ownership groups, built from your match-analysis deck and model documentation, so every member can own one model group and know it cold.

1. Assignments are set — see **Assignments** just below for who owns what and the core things each person needs to learn.
2. Each group's own section further down gives the full depth: what it owns, how it works, headline numbers, a worked example, self-test Q&A, and presentation talking points.
3. Everyone should also skim the **whole-system Q&A** near the end — a panel is as likely to ask across groups as within one.

Two of the five groups each carry one of the project's only two genuinely *trained* models (the CLIP identity model and the T-DEED action spotter) — those two owners have a training recipe to defend as well as a method.

Assignments

Member	Group	Core model(s)
Hadi	A — Detection & Tracking	YOLOv8x detectors, BoT-SORT, cut rule
Fazil	B — Pitch Mapping & Spatial Grounding	HRNetV2-W48 homography, team K-means, ball path
Aamir	C — Player Identity	ResNet-34 + PARSeq, CLIP identity model (trained)
Hasif	D — Events	T-DEED (trained), event-rule cascade
Danish	E — Statistics & Evaluation	distance/speed/sprint rules, evaluation methodology

What each person needs to learn, condensed to the essentials — full depth is in each group's own section below.

Hadi — Group A

- How YOLOv8x + BoT-SORT + the cut detector fit together: detect → track with cut-restart → join into segments.
- Headline numbers: 89.7% detection, 97.1% segment purity, cuts found 115/133 vs. 19 for the old rule.
- Why the ball detector is "the weakest link" (47.2%/72.1%), and why a fine-tuned version was tried but not adopted.
- The ~7.8-second-per-cut fact, and why it matters for everything downstream.
- No training recipe here — focus on *why* each pretrained or rule-based choice was made.

Fazil — Group B

- The three-part homography acceptance test (inliers, reprojection error, line agreement) and why cleaning matters.
- Headline numbers: 50.6% usable pitch view, jitter 1.1m→0.18m after cleaning, ball found 52.9% in the demo match.
- How teams are assigned by colour clustering, not numbers — and why grass gets masked out first.
- The ball-path search's skip-cost vs. new-piece-cost trade-off.
- Be ready for this one: team *names* are read by a person off the scoreboard, not detected by the system.

Aamir — Group C

- The identity model's architecture (CLIP ViT-B/16, 4 heads) and its full training recipe — one of only 2 trained models, expect to be asked in depth.
- The single biggest decision: "learn digits only from legible thumbnails" (74%→88%).
- The identity funnel — 89.7%→55.1%→52.3% named, 95% precision — and why that gap is coverage, not accuracy.
- Why the separate re-identification head was retired once this model existed.
- 19→48 training games raised accuracy meaningfully — know the before/after numbers.

Hasif — Group D

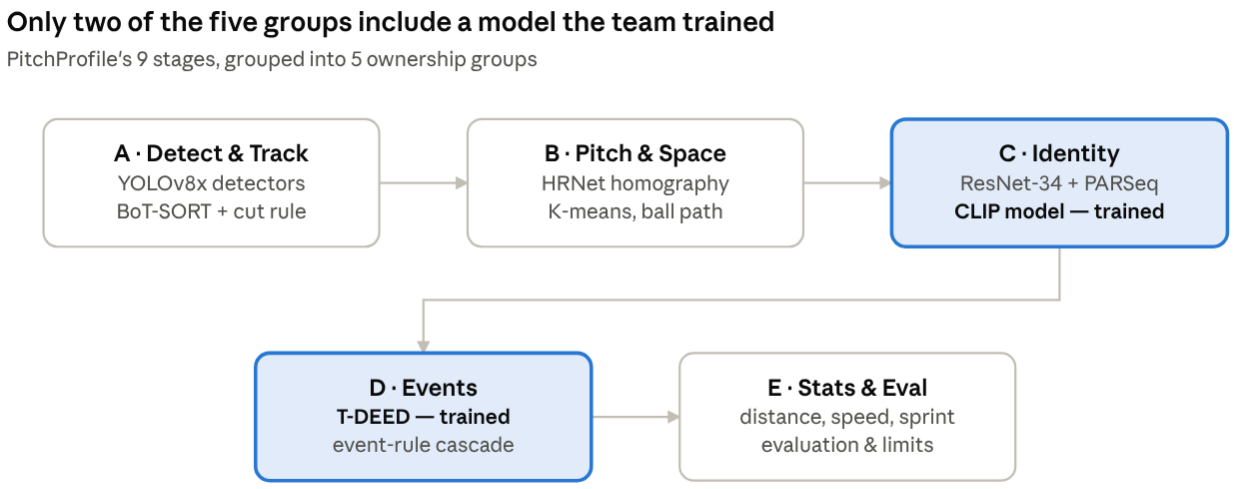
- T-DEED's architecture and training recipe — the other of only 2 trained models.
- The rule cascade's key constants (touch, spell, pass, carry, pressure) and the events with no rule fallback or an abandoned rule (tackle, take-on, shot).

- Why tackles are the weakest stat in the system (AP 0.06–0.10) — and that it's stated honestly, not hidden.
- The two worked examples: the t=2080s pass and the t=1021s shot.
- Event-to-player crediting is weak even with perfect identity (59%) — a likely question.

Danish — Group E

- The full measurement pipeline: smoothing → stepping → 11 m/s error-rejection → zones/sprints.
- The three rate-normalisation conventions and when each applies — expect "why three?"
- The demo match's team-totals table, and what it does and doesn't claim about the real match.
- The project's own "honest summary" paragraph — be ready to recite the gist.
- Why distance is a lower bound, not a true total.

The pipeline at a glance



PitchProfile's 9 stages, grouped into the 5 ownership groups

Groups C and D are the two shaded boxes — the project's only fine-tuned models. A, B and E run on pretrained weights or hand-written rules throughout.

Group A — Detection & Tracking

Owner: Hadi

Owns: YOLOv8x football detector, YOLOv8x ball detector (second, ball-only pass), BoT-SORT tracker + segment-joining, the camera-cut detector. All pretrained-as-released or hand-written rules — no training recipe in this group.

How it works

1. The video is sampled at 12.5 Hz (every 2nd frame of 25fps).
2. YOLOv8x (1,280px input, confidence floor 0.1, NMS IoU 0.5) finds player/referee/goalkeeper boxes. A second, ball-only YOLOv8x runs whenever the main detector finds no ball.
3. BoT-SORT tracks boxes frame to frame (Kalman filter + optical-flow camera-motion correction, two-pass matching at confidence ≥0.25 then ≥0.1, match threshold 0.85). Appearance re-identification is switched **off** inside the tracker — identity is handled later, by Group C.
4. The tracker restarts at every camera cut. Cuts are caught not by a model but by comparing 160×90px hue-saturation colour histograms between frames — a Bhattacharyya distance > 0.35 means a cut.
5. Short tracks are stitched into longer "segments" by a motion rule: gap ⩽ 1.2s and distance ⩽ 1.8m + 7.5m/s × that gap.

Numbers to know cold

- **89.7%** of visible player time detected (IoU ⩾ 0.5, measured on 2.03M FOOTPASS samples) — but only **63.6%** for tall boxes (>150px, close-ups) vs **88–92%** for the normal 60–150px range.
- Ball detector: **47.2% recall / 72.1% precision** at confidence 0.1 — called "the weakest link of the pipeline" in the team's own documentation.
- Cut detector found **115 of 133** labelled hard cuts on 42 minutes of SoccerNet video, against just **19** for an earlier mean-grey-level rule.
- **97.1%** segment purity (a segment rarely mixes two real players); **66.7%** of visible time sits in segments ⩾10s long; only **2.8%** of time has no segment at all.
- Demo match (95:20): **730 camera cuts** — about one every **7.8 seconds**.
- A fine-tuned ball detector was tried and *not adopted* (Appendix D.3): it found more balls alone (58.8% vs 47.2%) but, in the full pipeline, credited **fewer** passes to the right player (26.2% vs 34.2%) — so the untouched default detector stayed.

Worked example

730 cuts over 95:20 (5,720s) of the demo match works out to one cut roughly every **7.8 seconds**. That single number is why almost nothing downstream can assume a continuous camera view lasts long — tracking and identity both have to survive a cut about every 8 seconds, on average.

Self-test — cover the answers and quiz yourself

1. **Q:** Why run the detector with a confidence floor as low as 0.1 instead of something stricter? **A:** A low floor deliberately over-generates candidates; missing a real player costs more than a few false ones, which get cleaned up downstream (NMS, tracking, event logic).
2. **Q:** Why does the ball get its own separate detector instead of being one more class on the main one? **A:** The ball is tiny (often ~8px across) and rare; a dedicated ball-only pass, run only when the main detector finds nothing, improves recall without doubling cost every frame.
3. **Q:** What handles player identity inside the tracker itself? **A:** Nothing — appearance re-identification is deliberately switched off inside BoT-SORT; identity is entirely Group C's job, downstream.
4. **Q:** Why a hand-written rule for camera cuts instead of a trained model? **A:** It's cheap and measurably good (115/133 labelled cuts found) — no need to train a model for a problem a histogram-distance rule already solves well.
5. **Q:** What two numbers decide whether two short tracks get joined into one segment? **A:** Gap ⩽ 1.2s, and distance ⩽ 1.8m + 7.5m/s × that gap — consistent with how far a player could plausibly have run.
6. **Q:** Is the 89.7% detection figure measured over all broadcast video, or only a usable view? **A:** Only visible player time in a calibrated, usable view — not a claim about every raw frame.
7. **Q:** Why does accuracy drop so much for tall (>150px) boxes? **A:** Those are players very close to the camera — close-ups and replays — a less typical, more occluded framing than normal broadcast play.
8. **Q:** The team tried fine-tuning the ball detector. Why wasn't it adopted? **A:** It helped ball recall alone, but hurt the full pipeline — fewer passes ended up credited to the right player (26.2% vs 34.2%), possibly because its 720p training clips didn't match the pipeline's 1080p footage.
9. **Q:** What is this group's single most honest weak point? **A:** Ball detection — called "the weakest link of the pipeline" in the documentation itself.
10. **Q:** What's the practical effect of a camera cut every ~8 seconds? **A:** Every track, segment and identity has to be re-established constantly; nothing can assume a long, uninterrupted view of a player.

Presentation talking points

The problem: Broadcast football video cuts camera angle constantly, and the ball is tiny and often hidden — just *seeing* the match reliably is already hard. **Our answer:** A proven pretrained detector (YOLOv8x) run with a deliberately low confidence floor, a motion-

aware tracker (BoT-SORT) that survives most cuts, and a lightweight colour-histogram rule that catches the cuts themselves. **How it works:** Detect → track with cut-restart → join short tracks into longer segments. **Outcome:** 89.7% of visible player time detected; 97.1% segment purity; cuts caught 6× better than the rule it replaced. Ball detection stays the honest weak point — say so plainly if asked.

Group B — Pitch Mapping & Spatial Grounding

Owner: Fazil

Owns: HRNetV2-W48 pitch-keypoint model + RANSAC homography + multi-stage cleaning, team assignment (K-means on shirt colour) + direction/goalkeeper rules, the ball-path dynamic-programming search. Pretrained keypoint network plus hand-written methods — no training recipe in this group.

How it works

1. HRNetV2-W48 predicts 57 known pitch-point heatmaps from a 960×540 input (confidence floor 0.1486).
2. RANSAC fits a homography (the 3×3 matrix mapping pitch ↔ image) — accepted only with ≥5 inliers within 6px, reprojection error <4px, and agreement with the painted pitch lines.
3. Multi-stage cleaning: median-of-7-neighbours outlier rejection (>1m), gap interpolation (≤1.2s), 5-sample smoothing, and a 12m-jump heuristic that flags a likely undetected cut.
4. Teams: K-means (2 clusters, 12 random starts, sqrt-count-weighted, lightness half-weighted) on CIELAB torso colour from the central 64%×15–50% of each box, grass masked out in HSV.
5. Direction: the team with the smaller mean x-position defends the left goal (gap ≥1.5m = "inferred"); goalkeepers get the detector's own label + nearest-own-goal assignment + the identity model's GK head as a second opinion.
6. Ball path: a per-shot dynamic-programming search over ball candidates (cost −0.5·ln(confidence), skip cost 1.0, new-piece cost 4.0, motion limits ≤90px/sample and <38m/s implied speed); gaps ≤8 samples are bridged inside the search, ≤0.8s interpolated after (marked, never used as touch evidence).

Numbers to know cold

- **57** pitch keypoints; cleaning drops jitter from 1.1m→0.18m (90th percentile) and 9–18m→0.5m (99th percentile), measured on Chelsea v Burnley.

- Demo match: **50.6%** usable pitch view — just over half the match has reliable coordinates at all.
- Ball located in a **median of 64%** of live play across 22 halves (range 47–80%); demo match **52.9%**.
- Goalkeeper direction called right on **8/8** checkable halves.
- The demo match's team *names* (Chelsea / Man Utd) came from a person reading the broadcast scoreboard — the system itself only ever outputs "Team A / Team B" by colour.

Worked example

A homography is accepted only when it clears **three** checks at once: ≥5 inlier keypoints within 6px, reprojection error under 4px, and agreement with where the lines are actually painted. On the demo match, that combination of checks passes only **50.6%** of the time — meaning roughly half the match has no trustworthy pitch coordinates at all. That's why every rate in the report is normalised to *visible time*, not total match time: the other half simply isn't measured, not assumed to be zero.

Self-test — cover the answers and quiz yourself

1. **Q:** Why 57 keypoints instead of just the 4 pitch corners? **A:** More points make the fit robust to partial views (most broadcast shots show only part of the pitch) and give RANSAC enough inliers to reject outliers.
2. **Q:** What three conditions make a homography "accepted"? **A:** ≥5 inliers within 6px, reprojection error <4px, agreement with the painted lines.
3. **Q:** Why treat a sudden 12m position jump as "probably an undetected cut" rather than real motion? **A:** No player can move 12m between samples at 12.5Hz — it's a calibration failure, most likely from a camera cut the cut-detector missed.
4. **Q:** Why cluster shirt *colour* for teams instead of reading kit numbers? **A:** It's cheap, needs no labels, and works on nearly every frame — colour is visible almost continuously, legible numbers are not.
5. **Q:** Why mask out grass before clustering torso colour? **A:** Visible grass around a player would bias the two clusters toward "grass vs not-grass" instead of "team A vs team B."
6. **Q:** How is playing direction decided, and why is it sometimes only "inferred"? **A:** The team with the smaller mean x-position is assumed to defend the left goal; it's labelled "inferred" unless the two teams' mean x differ by ≥1.5m.
7. **Q:** Why does the ball-path search have both a "skip" cost and a "new-piece" cost? **A:** They trade off two failure modes — skip lets the path ignore one bad detection without breaking; the pricier new-piece cost discourages restarting the trajectory from scratch.

8. **Q:** What share of live play is the ball actually located in during the demo match, and is that typical? **A:** 52.9% — on the low end of the 47–80% range (median 64%) seen across 22 halves.
9. **Q:** How did the system know to call the two teams "Chelsea" and "Manchester United"? **A:** It didn't — a person read the names off the broadcast scoreboard; the pipeline itself only distinguishes Team A from Team B by colour.
10. **Q:** What happens to statistics in the ~49% of the match with no usable pitch view? **A:** Nothing is computed for that time — no distance, speed, position or event — it's missing, not estimated or interpolated across.

Presentation talking points

The problem: Turning pixel boxes into real distances and speeds means knowing exactly where the camera is pointed at every moment — and broadcast cameras move constantly and rarely show the whole pitch. **Our answer:** A pretrained keypoint network finds known pitch markings, RANSAC fits a calibration only when it's reliable, and several cleaning passes smooth the result; team colour and ball path are recovered with classical, hand-tuned methods — no training needed for either. **How it works:** Detect pitch keypoints → fit & accept/reject a homography → clean the signal → assign teams by colour → search for the ball's path. **Outcome:** 50.6% of the demo match has a usable pitch view (90th-percentile jitter under 20cm after cleaning); the ball is found just over half the time. Everything downstream — distance, speed, events — only exists where this group's calibration succeeds.

Group C — Player Identity

Owner: Aamir

Owns: ResNet-34 legibility classifier + PARSeq number reader (pretrained), and the team's own **fine-tuned CLIP ViT-B/16 identity model** — one of only two models the team actually trained — plus cross-cut identity-grouping rules.

How it works

1. ResNet-34 classifies whether a thumbnail's shirt number is legible; PARSeq reads the digits where it is.
2. The identity model: a CLIP ViT-B/16 backbone over a 224×224 thumbnail, with 4 heads off its 768-dim feature — tens-digit (10 classes), units-digit (10 classes), a 256-dim normalised contrastive appearance embedding, and a goalkeeper-probability head.

3. Inference: up to 24 thumbnails per track; confidence-weighted digit voting using only thumbnails with legibility ≥0.5; unnumbered segments are grouped by appearance (never joining two segments seen on screen at once, up to 15s tolerance); a group is named once one number holds ≥85% of its votes; gaps <5s are bridged by position; GK calls need a team-typical-score >0.3 (or exclusion above 15% of a team's detections) *and* being within 16.5m of the goal line.

Training recipe — the richest training story in the project

- **48 FOOTPASS games, 6.0M thumbnails, 1,485 players.** Batches of 16 players × 4 thumbnails, guaranteeing hard-negative teammates (similar kit) in every batch.
- Augmentation: random resize 45–100% to simulate 720p blur — **never mirrored**, since digits aren't mirror-symmetric.
- Loss (3 terms): digit cross-entropy with 0.05 label smoothing; 256-dim supervised-contrastive appearance loss at temperature 0.1; 0.5-weighted BCE goalkeeper loss.
- AdamW: LR 1e-5 backbone / 2e-4 heads, weight decay 0.05, 300-step warmup, cosine decay, bfloat16, grad-clip 1.0, **10,000 steps on one RTX 3090 (24GB)**.
- **The single most important design decision: "learn digits only from legible thumbnails."** Training the digit heads only where the number is actually readable raised legible-vote-right-number from **74%→88%** overall, and **42–53%→73%** on the hardest, most-ambiguous-kit game.
- Version history: v1 (19 games, all crops) → v2 (19 games, legible-only, 6,000 steps) → adopted (48 games). Going 19→48 games raised right-name share **48.9%→53.0%** on FOOTPASS and **71.8%→75.7%** on SoccerNet.

Numbers to know cold

- Identity funnel: **89.7%** detected → **55.1%** → **52.3%** of visible time named right — and **95%** of names given are correct.
- Confident (≥0.5 legibility) number reader: **94.2%** right while reading **38%** of tracklets.
- Goalkeeper head: **89.9%** recall / **99.0%** precision; appearance matching **98.3%** (3 held-out FOOTPASS games, 13,610 tracklets, 406,205 crops).
- ResNet-34+PARSeq alone: **80.3%** of 1,211 SoccerNet-Jersey-2023 tracklets right (incl. "not visible" as correct); **76.8%** of tracklets with a genuinely readable number read correctly.
- A separate re-identification head was tried and *retired* (Appendix D.2): it matched only 71.6–90.1% of held-out segments vs. this model's 98.3% — dropped once the identity model existed.
- Untested extra: connecting the real match line-up raised named-right share **53.0%→64.2%** in one isolated test — not used in any reported pipeline figure.

Worked example

The identity funnel, read as a story: of all visible player-time, **89.7%** is detected at all. Of that, **55.1%** ends up grouped into something the system is willing to consider naming. Of *all* visible time, only **52.3%** ends up carrying a name — but when it commits to one, it's right **95%** of the time. The system is deliberately conservative: it would rather stay silent than guess wrong.

Self-test — cover the answers and quiz yourself

1. **Q:** Why train a bespoke identity model instead of just using the pretrained number reader? **A:** The number reader only works when a number is clearly legible — it can't carry a name across gaps, cuts or blur; the identity model adds appearance matching and cross-cut grouping on top.
2. **Q:** What's the single biggest design decision in training this model, and what did it buy? **A:** "Learn digits only from legible thumbnails" — raised legible-vote-right-number from 74%→88% overall (42–53%→73% on the hardest kit game).
3. **Q:** Why batch 16 players × 4 thumbnails instead of random sampling? **A:** It guarantees hard negatives — similar-looking teammates — in every batch, exactly the confusion the appearance embedding needs to resolve.
4. **Q:** Why is the resize augmentation never mirrored? **A:** Shirt numbers aren't mirror-symmetric; flipping would teach the model to read digits backwards.
5. **Q:** What are the identity model's 4 heads? **A:** Tens-digit, units-digit (10 classes each), a 256-dim contrastive appearance embedding, and a goalkeeper-probability head — all off one shared CLIP feature.
6. **Q:** The model names only 52.3% of visible time but is 95% accurate when it does — what does that say about its design philosophy? **A:** It's deliberately conservative: coverage, not correctness, is the limiting factor — it would rather stay silent than commit to a likely-wrong name.
7. **Q:** Why was the separate re-identification head retired? **A:** The identity model's own appearance embedding matched held-out segments far better (98.3% vs 71.6–90.1%) — no reason to keep a weaker component doing the same job.
8. **Q:** What stops a non-goalkeeper from being tagged as GK? **A:** A team-typical-score threshold (>0.3, or exclusion above 15% of a team's detections) plus a geometric check — within 16.5m of the goal line.
9. **Q:** Going from 19 to 48 training games raised accuracy by how much? **A:** Right-name share rose 48.9%→53.0% on FOOTPASS and 71.8%→75.7% on SoccerNet.
10. **Q:** What untested feature could push named-right share higher, and by how much in one isolated test? **A:** Connecting the real match line-up — 53.0%→64.2% in one isolated test — but it's not used in any reported pipeline figure, since it was only tested

with true squads.

Presentation talking points

The problem: A player's identity has to survive dozens of camera cuts and blurry frames, and shirt numbers are often simply not visible — get it wrong and every stat attaches to the wrong person. **Our answer:** Fine-tune a CLIP model to read shirt digits and recognise appearance together, trained to trust digits only where it can actually see them clearly — then name conservatively, favouring "unnamed" over "wrong." **How it works:** Read legible numbers with confidence-weighted voting → group unnumbered segments by appearance → name a group once one number dominates its votes → bridge short gaps by position. **Outcome:** 52.3% of visible time named, 95% of those names correct — "partial but trustworthy when given" is the line to use if asked why it isn't closer to 100%.

Group D — Events

Owner: Hasif

Owns: the **fine-tuned T-DEED action spotter** — the other of only two models the team trained — plus the full hand-written event-rule cascade (touch, spell, pass, carry, interception, tackle, take-on, shot, cross, pressure).

How it works

1. **T-DEED:** RegNet-Y 200MF with gate-shift modules (per-frame features) + a temporal encoder-decoder (sharpens frame-to-frame differences) + 3 heads (12-class action, sub-frame timing offset, team side). Runs on 8s/100-frame clips (12.5fps, 796×448px, 3/4 overlap), merged with soft-NMS (12-frame window), score floor 0.05. Tackle/block team assignment uses a *previous-touch* heuristic instead of the model's own side output (78–97% right vs. 65–78%). Statistics are **expected counts** (summed scores), not thresholded counts.
2. **Event rules**, built on top of Group A/B's positions and ball path: touches from ball-velocity spikes (`TOUCH_DV=4m/s` , `TOUCH_RADIUS_M=2.0m`); possession spells (`SPELL_MAX_GAP_S=2.5s`); passes (`MAX_PASS_FLIGHT_S=4s` , ≤35m/s implied speed, progressive/long/cross flags via `PROGRESSIVE_M=10m` / `LONG_PASS_M=30m`); carries (`CARRY_MIN_M=2.0m`); interceptions (`INTERCEPT_MIN_SPEED=6m/s`); take-ons (ball-retention decides the outcome); pressure (`PRESSURE_RADIUS_M=3.0m`).
3. **Tackles have no rule** — the rule-based attempt only reached 6–14% precision, so tackles come entirely from T-DEED's expected counts. **Shots have no rule either**, by design — decided entirely by the video model.

Training recipe

- Starts from the **published SoccerNet-2025 T-DEED checkpoint**; backbone stem/s1/s2 frozen (16K/3.2M params) to fit a **single RTX 3090 (24GB)**.
- Trained on 5 SoccerNet-BAS + 48 FOOTPASS (partial-label) games.
- Devkit recipe: mixup Beta(0.2,0.2), colour/blur augmentation, 5× foreground-class weighting, class + displacement + 2× team loss.
- Project-specific changes: a partial-label cross-entropy variant, a lower fine-tuning LR, extra tackle/block weighting, rare-action-centred clip sampling.
- **3,000 steps, ~3 hours.** Best checkpoint chosen by held-out class loss; loss fell 3.39→1.80.
- A second, more tackle-weighted fine-tune was tried and **not adopted** (Appendix D.5): tackle AP only 0.06→0.08 on FOOTPASS and 0.10→0.09 on BAS, crosses and headers got worse.

Numbers to know cold

- Passes/drives AP **0.91 / 0.89** on FOOTPASS validation; tackle AP weak at **0.06–0.10**.
- Published→fine-tuned mean AP: **0.606→0.628** SoccerNet-BAS, **0.478→0.604** FOOTPASS — fine-tuning helped far more on FOOTPASS.
- Shot spotting: **0.66/0.68** precision/recall on train+val halves, **0.68/0.74** on 4 held-out test halves.
- Rule-based events vs. annotations on perfect positions (SoccerTrack, mean of two halves): pass (right passer) **~0.68/0.70**; pass (any passer) **~0.75/0.77**; carry vs. "Drive" **~0.65/0.56**; cross **~0.51/0.17**; tackle (rules) **0.06–0.14**; block (rules) **0–0.10**.
- Per-player pass counts correlate with annotations at **0.89 Pearson** (23 players, 0.91 Spearman) — why players can still be compared on passing despite individual errors.
- **Weakest link:** even with perfect identities, only **59%** of labelled passes land on the right player's track; about **1 in 4** spotted passes/carries can't be credited to anyone.

Worked example

At **t=2080s** of the demo match, the rules correctly flagged a completed **progressive pass**: 1.52 seconds of flight, 22.7m travelled, 18.4m of it forward, at an implied speed of ~14.9 m/s — comfortably inside the 4s/35m/s pass limits and past the 10m progressive threshold. At **t=1021s**, a shot from 21.2m out was called — entirely by the video spotter, with no rule involved at all, since shots have no rule-based fallback.

Self-test — cover the answers and quiz yourself

1. **Q:** Why do tackles come from the video spotter instead of a rule, when every other ball event has one? **A:** The rule-based attempt only reached 6–14% precision / 8–12% recall — too unreliable; the trained spotter (AP 0.06–0.10) was still weak but the better of two bad options.
2. **Q:** Why are tackle/block reported as "expected counts" rather than simple thresholded counts? **A:** The spotter's scores behave like probabilities; summing them gives a steadier estimate from a weak detector, flagged with a dagger to show it's a sum of scores, not a hard count.
3. **Q:** Why does shot detection have no rule-based fallback at all? **A:** It's a deliberate choice — the video model alone decides shots; earlier rule-based and classifier attempts were both worse (see the retired shot classifier).
4. **Q:** What happened to the original take-on/dribble detection approach, and why? **A:** "Following the defender" was discarded — it flipped its call too often on ~5cm of position noise; replaced by a rule where ball retention alone decides the outcome.
5. **Q:** What's the single weakest link in the whole events chain, even with perfect identity? **A:** Event-to-player crediting — only 59% of labelled passes land on the right player's track, and about a quarter of spotted passes/carries can't be credited to anyone.
6. **Q:** Why did fine-tuning help far more on FOOTPASS (0.478→0.604) than SoccerNet-BAS (0.606→0.628)? **A:** FOOTPASS is this project's own data distribution; the published checkpoint was never tuned to its specific broadcast style, so there was more room to improve.
7. **Q:** Why freeze part of the backbone during fine-tuning? **A:** To fit the whole job on one 24GB RTX 3090 — freezing early layers keeps memory and compute down while heads and later layers still adapt.
8. **Q:** A second, tackle-weighted fine-tune was tried. Why wasn't it adopted? **A:** It barely moved tackle AP (0.06→0.08 FOOTPASS) and made crosses and headers worse.
9. **Q:** Individual passes are sometimes wrong, so why can players still be fairly compared on passing? **A:** Per-player pass counts correlate with ground truth at 0.89 Pearson — errors wash out in aggregate.
10. **Q:** Give a concrete example of a correctly-flagged progressive pass. **A:** t=2080s of the demo match: 1.52s, 22.7m total, 18.4m forward, ~14.9 m/s.

Presentation talking points

The problem: "What happened" is harder than "where is everyone" — some actions (tackles, headers) are nearly invisible to rules built only on positions. **Our answer:** Fine-tune a video action-spotter (T-DEED) for actions that truly need to see the video, and

layer a careful, constants-driven rule cascade on positions for everything else — abandoning rules that didn't work (tackles) rather than keep shipping a bad one. **How it works:** Detect touches from ball-velocity spikes → group into possession spells → classify passes/carries/interceptions/take-ons from motion → let the video model alone decide shots and tackles. **Outcome:** Passes/drives reach AP 0.91/0.89; per-player pass counts correlate with ground truth at 0.89. Tackles and player-crediting stay the honest weak points — know those numbers if asked "why not higher?"

Group E — Statistics & Evaluation

Owner: Danish

Owns: distance/speed/sprint/heatmap computation, the three rate-normalisation conventions, and the project's overall evaluation methodology and limitations. This owner is the one who can defend what every number does and doesn't mean.

How it works

1. Box positions reach the pitch via Group B's homography, using the foot point (bottom-centre of the box), with up to 3m outside the painted lines still accepted.
2. Smoothing-then-distance: a ~1-second (13-sample) smoothing window, then speed computed in 0.4s steps; any step implying >11 m/s is discarded as noise.
3. Speed zones (km/h): Walk / Jog / Run / High-speed / Sprint at **7.2 / 14.4 / 19.8 / 25.2** cutoffs.
4. **Sprint** = ≥25.2 km/h sustained for ≥3 consecutive steps (~1s), counted once per burst.
5. A high-intensity-run taxonomy (forward run, run-in-behind, overlap/underlap, pressing run, recovery run, etc.) with its own geometric thresholds (e.g. ≥5.5 m/s sustained ≥1.0s).
6. Heatmaps: a 20×32 normalised grid, attack-flipped so every player's territory reads the same direction.
7. **Three rate conventions**, used for different questions: per-90-on-screen (the default), per-90-team-live-play, and per-100-touches — plus "expected counts, shown with a dagger" for anything that comes from a model's summed scores rather than a hard count.
8. Whole-match merging shifts the second half by 2,700s and mirrors direction, since teams swap ends at half-time.

Numbers to know cold

- Demo match team totals: **Man Utd** 55.6% possession, 393 touches, 489 passes (75.9% complete), 23 shots, 33.5km distance. **Chelsea** 44.4%, 341 touches, 443 passes (67.7%), 14 shots, 31.5km.
- Distance is explicitly a **lower bound** — it sums only on-screen time: "a midfielder who covered 11km may show 3km if the camera showed him a quarter of the time."
- Only **50.6%** of the demo match has a usable pitch view at all (Group B's number — know it, since it caps everything this group computes).
- Step speeds above **11 m/s** are discarded outright as tracking noise, not real motion.
- Evidence base: **22 analysed halves of 11 matches**; ground truth from one fully-annotated match (SoccerTrack 117092) plus two SoccerNet-NDA datasets.
- Throughput: ~22–25 sampled frames/s on an RTX 3090; the demo match took 3,044s of detection time for both halves.
- **The project's own "honest summary":** *"Passes, carries, distance and speed are reliable where the player is on screen. Shots are reasonable. Identity and credit are good enough for per-player comparison with care. Tackles and blocks are rough. The report is a view of the broadcast, not of the whole match."*

Worked example

An unnamed Chelsea player ("B-Y2") is tracked for 26 seconds. Smoothed, stepped positions sum to **89m** of distance. Top speed reaches **26.2 km/h** — above the 25.2 km/h sprint cutoff — sustained long enough to count as exactly **1 sprint** (a single burst, not one count per fast step). This is the whole pipeline in miniature: raw positions → smoothing → stepped speed → zone/sprint classification.

Self-test — cover the answers and quiz yourself

1. **Q:** Why is "distance" a lower bound rather than a true total? **A:** It only sums time the player was actually on screen in a calibrated view — time off-screen isn't estimated, it's simply missing.
2. **Q:** Why does the report need three different rate conventions instead of one "per-90"? **A:** Each answers a different question — on-screen time controls for how much of a player you saw; team-live-play time controls for how much the ball was in play; per-100-touches compares efficiency rather than volume.
3. **Q:** What happens to a calculated step speed above 11 m/s? **A:** It's discarded outright as noise — no real player sustains that speed, so it's treated as a tracking glitch.
4. **Q:** What exactly counts as one "sprint"? **A:** ⩾25.2 km/h sustained for at least 3 consecutive 0.4s steps (~1 second), counted once per sustained burst.

5. **Q:** Why smooth position over ~1 second before computing speed, rather than differencing raw samples? **A:** Raw per-sample positions are noisy; smoothing first gives a far more stable speed estimate.
6. **Q:** Why isn't the whole match just the two halves concatenated? **A:** The second half needs its time shifted by 2,700s and its direction mirrored, since teams swap ends at half-time.
7. **Q:** In the demo match, Man Utd had more possession, passes and completion than Chelsea. Does that describe the real match or the broadcast? **A:** Only what the broadcast showed and the pipeline measured — "a view of the broadcast, not of the whole match."
8. **Q:** What's the project's own one-paragraph honest summary of reliability? **A:** Passes/carries/distance/speed reliable on-screen; shots reasonable; identity and credit good enough for per-player comparison with care; tackles and blocks rough.
9. **Q:** How much evidence underlies these numbers, and why does that matter? **A:** 22 analysed halves of 11 matches, with ground truth from one fully-annotated match plus two NDA datasets — small enough that results on other broadcasters or camera styles may differ.
10. **Q:** Why might two "per-90" stats for the same player disagree between two reports? **A:** "Per-90" needs a stated convention — on-screen minutes, team-live-play minutes and per-100-touches are different denominators that are easy to conflate if unlabelled.

Presentation talking points

The problem: Every number here is a measurement with error, not ground truth — someone has to be able to say exactly how and why a given stat could be wrong. **Our answer:** A disciplined pipeline (smooth → step → reject noise → classify into zones/sprints/runs) plus three clearly labelled rate conventions, validated against a small but real ground-truth set, with limitations stated plainly rather than hidden. **How it works:** Project positions to the pitch → smooth → compute stepped speed → classify into zones/sprints/runs → normalise by the denominator the question actually needs. **Outcome:** Team totals and rate-based comparisons are reliable; raw player totals are a lower bound. This owner can say, convincingly, exactly what any given number does and doesn't mean.

Whole-system Q&A (everyone should know these)

Every member should be able to answer these, regardless of which group they own — a panel is as likely to probe across groups as within one.

1. **Q:** Across the whole system, how many models did the team actually *train*? **A:** Only two — the CLIP ViT-B/16 identity model (Group C) and the T-DEED action spotter (Group D). Everything else is pretrained-as-released weights or hand-written rules.
2. **Q:** Why is broadcast football video fundamentally hard to analyse? **A:** The camera cuts constantly (about every 8 seconds in the demo match); the ball is found less than half the time; only about half of player time ties to a known shirt number.
3. **Q:** What happened to the project's original plan to rate players' archetypes / playing styles? **A:** Removed on 8 October 2026 — no player was ever labelled, so the planned approach was never even fitted. It's now listed only as "a possible extension, not a result."
4. **Q:** What was the "short-clip analysis" path, and why was it removed? **A:** An earlier version let users upload short clips, calibrate by hand, and track with a lighter detector — it used none of the identity, event or action-spotting models, so it was dropped in favour of full-half, full-match analysis only.
5. **Q:** Why are team-level stats more reliable than player-level stats? **A:** Identification is only ~52% complete (though 95% accurate when given), and event-to-player crediting is weak (~59% of passes) — individual errors partly cancel out in team totals but not for any one player.
6. **Q:** What does the system explicitly NOT determine about a shot? **A:** Whether it was on target, blocked, or a goal — it only detects that a shot happened; outcome needs a human reviewer.
7. **Q:** Why does the ball effectively have "no height," and what does that break? **A:** Ball positions are projected onto the flat pitch from 2D video, so a lofted ball is placed where it isn't in 3D — making aerial passes and headers the least reliable event types.
8. **Q:** Give one example where the team fine-tuned something and specifically chose *not* to adopt it. **A:** A fine-tuned ball detector found more balls (58.8% vs 47.2%) but credited fewer passes correctly end-to-end (26.2% vs 34.2%) — so the untouched default detector was kept.
9. **Q:** Why do several components carry non-commercial or copyleft licences, and what does that mean practically? **A:** YOLOv8x (AGPL-3.0), HRNetV2-W48 (GPL-2.0), ResNet-34+PARSeq (CC BY-NC/Apache-2.0 mix), T-DEED's starting checkpoint (GPL-3.0) — fine for research/coursework, but commercial use would need a licensing review first.
10. **Q:** What hardware trained both fine-tuned models? **A:** A single RTX 3090 (24GB) — the identity model for 10,000 steps, the action spotter for 3,000 steps (~3 hours).
11. **Q:** How does the project itself frame what this system is, versus commercial alternatives? **A:** "An open research prototype" — broadcast tracking is already sold commercially (e.g. by SkillCorner); this complements rather than replaces that, for now.

12. **Q:** If asked "why not just train one end-to-end model for everything?" — what's the honest answer? **A:** The sub-problems (seeing players, finding the ball, naming players, spotting rare actions) have very different data availability and failure modes; decomposing into pretrained detectors + two targeted fine-tunes + transparent rules made every piece measurable and explainable, at the cost of errors compounding downstream.

Sources

This guide is built entirely from the two files provided:

- **PitchProfile_Match_Analysis.pdf** — the 21-slide deck: motivation, the 9-stage pipeline, per-stage results, and the scale-up roadmap.
- **PitchProfile_Model_Documentation.docx** (8 October 2026) — the full technical write-up: architecture, training recipes, every threshold and constant, complete results tables, and the retired/removed appendix.

Every number and quote above traces back to one of these two files.